

The zero-spacing limit for any plaquette action and any dimension

Result, 2026-09-27. The series/parallel note worked with the heat-kernel action. Here the question is how the $a \rightarrow 0$ limit depends on the choice of plaquette action, and on the dimension.

1. **Two semigroups.** A series move convolves plaquette weights, so it multiplies their character coefficients; heat kernels form a family closed under it. A parallel move multiplies weights pointwise, so it adds their logarithms; exponential families $e^{-\beta V}$ with a fixed potential (for example Wilson) are closed under it. In the small-field regime the two agree: for $U(1)$ the product of two heat kernels is exactly a heat kernel at the parallel heat time $st/(s+t)$ plus shifted terms whose mixture weights are $O(e^{-2\pi^2/(s+t)})$ (Proposition 1).
2. **Two dimensions, any action** (Theorem 2). The possible continuum limits under infinitesimal symmetric class-function plaquette refinement with finite representation exponents are exactly the symmetric conjugation-invariant Lévy exponents; for compact connected simple G they are $\psi(R) = \frac{\sigma_2}{2}C_2(R) + \int (1 - \operatorname{Re} \chi_R/d_R) d\nu$, with $\psi(R) = \lim a^{-2}(1 - \hat{c}_R(a))$. Yang–Mills is the Gaussian member, $\nu = 0$, selected by a Lindeberg condition on the action. A weight with an atom at a centre element produces a centre-vortex gas, and the circle spectrum becomes $E_R - E_0 = \hbar c L \psi(R)$.
3. **Any dimension: an error budget per physical volume.** One refinement step leaves three kinds of error. The perturbative part is of relative order t_n , the plaquette heat time at step n . The candidate large-field bound scales as $a_n^{-D} e^{-c/t_n}$ per physical volume, with c the minimal large-field action of the chosen lattice action in units of the inverse heat time. The jump part is the action’s Lévy content. With $t \propto \lambda_D a^{4-D}$:
 - $D < 4$: the displayed perturbative and exponential scales are summable at fixed λ_D , provided uniform bounds and stability supply this error budget;
 - $D = 4$: the perturbative part resums into the running coupling, and the large-field part decays only as a power of a , $(a\Lambda)^{2b_0c}$, giving the sufficient strict power threshold $2b_0c > 4$, that is $c > 96\pi^2/(11N)$ for pure $SU(N)$: $c > 43.1$ for $SU(2)$, $c > 28.7$ for $SU(3)$, that is $12/(11N)$ of the instanton action. Instantons ($c = 8\pi^2 \approx 79$) pass for every N ; for lattice dislocations this is the known 6/11 criterion of Pugh–Teper and Gökeler et al. (1989);
 - $D > 4$: t grows as $a \rightarrow 0$ at fixed λ_D , so no small-field regime exists at short distances.

The exact two-dimensional classification and the higher-dimensional error budget have different status. For $D \geq 3$, uniform blocking, observable normalization, large-field entropy and reconstruction must still be controlled before asserting universality or equality of gaps. The three scales organize that task. The ingredients are established (character expansions, Poisson summation, Hunt’s classification of convolution semigroups, the one-loop running); the organization by the two semigroups and the error budget is the contribution, with no novelty claimed for the parts.

1. Series and parallel act on different coordinates of the weight

Let w be a class-function probability density on a compact connected Lie group G (Haar measure normalized), symmetric, $w(g^{-1}) = w(g)$, with character coefficients $\hat{c}_R = d_R^{-1} \int w \overline{\chi_R} dg \in [-1, 1]$, so that $w = \sum_R d_R \hat{c}_R \chi_R$. Group metric as in the series/parallel note §1 ($-\Delta \chi_R = C_2(R) \chi_R$).

- **Series.** Integrating out an edge shared by two faces convolves their weights, and $(w_1 * w_2)_R = \hat{c}_R(w_1) \hat{c}_R(w_2)$. The heat kernels k_t , $\hat{c}_R = e^{-tC_2(R)/2}$, form the one-parameter family closed under this product with t additive.

- **Parallel.** Two faces with the same holonomy combine by the pointwise product $w_1 w_2$, and $\log(w_1 w_2) = \log w_1 + \log w_2$. The families $w_\beta \propto e^{-\beta V}$ with fixed V are closed under this operation with β additive; Wilson's $V = 1 - \text{Re tr } U/N$ is one.

Proposition 1 (parallel product of $U(1)$ heat kernels). With $k_s(\theta) = \sum_{n \in \mathbb{Z}} e^{-sn^2/2} e^{in\theta}$ and $\tau = st/(s+t)$,

Correction (GPT-6 Astra referee), 2026-09-27 — REFINE: normalized Haar measure supplies a missing factor 2π ; the small weights are mixture weights, with a separate pointwise ratio.

$$k_s(\theta) k_t(\theta) = \sqrt{\frac{2\pi}{s+t}} \sum_{m \in \mathbb{Z}} e^{-2\pi^2 m^2/(s+t)} k_\tau\left(\theta + \frac{2\pi ms}{s+t}\right).$$

Proof. Poisson summation gives $k_s(\theta) = 2\pi \sum_w G_s(\theta + 2\pi w)$ with $G_s(x) = (2\pi s)^{-1/2} e^{-x^2/(2s)}$. For Gaussians, $G_s(x)G_t(y) = G_{s+t}(x-y) G_\tau((tx+sy)/(s+t))$. Put $x = \theta + 2\pi w$, $y = \theta + 2\pi w'$ and $m = w' - w$; the sum over w at fixed m is $k_\tau(\theta + 2\pi ms/(s+t))/(2\pi)$. Multiplying the two initial factors 2π leaves the prefactor $2\pi/\sqrt{2\pi(s+t)}$. $\square$

The $m = 0$ term is the heat kernel at the parallel heat time, the conductance rule of the series/parallel note §3. The terms $m \neq 0$ are shifted images, of relative mixture weight $e^{-2\pi^2 m^2/(s+t)}$. After normalizing the product to integrate to one, these weights give an exponentially small total-variation correction to k_τ . Pointwise, the relative correction also contains $k_\tau(\theta + 2\pi ms/(s+t))/k_\tau(\theta)$; it need not be small near the cut locus. For $s = t$ and $\theta = \pi$, a shifted term is of the same leading exponential order as the unshifted term. Heat kernels close exactly under series, and fixed-potential exponential families under parallel. Both reduce to the Gaussian in the small-field regime, where they agree.

A limit of validity for heat-kernel anisotropy. Equation (1) of the series/parallel note matches heat times to the continuum action in the small-field regime. Refining one direction alone, for instance Euclidean time at fixed spatial lattice, sends the transverse heat times to infinity. Heat-kernel weights then become flat up to corrections $e^{-tC_2/2}$, whereas the Kogut–Susskind limit needs a magnetic weight $e^{-a_0 V}$ with a_0 the time step. The Hamiltonian limit is reached within the exponential family. The directional halvings of an isotropic cycle change heat times only by factors of two, so the propositions of that note are unaffected.

Referee verdict, 2026-09-27: ACCEPT with scope. At fixed spatial spacing write $t_{jk} = B/a_0$, $B > 0$. On a connected group the first nonconstant eigenvalue gives $k_{B/a_0} = 1 + O(e^{-BC_{\min}/(2a_0)})$; dividing its logarithm by a_0 gives zero magnetic potential. Equation (1) remains a definition but this time-only heat-kernel family yields an electric Hamiltonian with vanishing magnetic term. A spatial weight $e^{-a_0 V}$, together with temporal heat kernels, restores the desired Hamiltonian. Other weights with that first-order expansion also work.

2. Two dimensions: the continuum limits of every plaquette action

In two dimensions every refinement is a series move (the series/parallel note, §2). A face of area A tiled by A/a^2 plaquettes of weight w_a has, after integrating the interior edges, the weight whose coefficients are $\hat{c}_R(a)^{A/a^2}$.

Correction (GPT-6 Astra referee), 2026-09-27 — REFINE: require finite exponents, use a general invariant diffusion tensor for nonsimple groups, and include the converse construction.

Theorem 2. Initially take G compact, connected and simple, and let $w_a \geq 0$ be normalized symmetric class-function plaquette weights on the lattice of spacing a , with $\hat{c}_R(a) \rightarrow 1$ for every R . Suppose that for every R the limit

$$\psi(R) = \lim_{a \rightarrow 0} \frac{1 - \hat{c}_R(a)}{a^2} = \lim_{a \rightarrow 0} a^{-2} \int_G \left(1 - \frac{\text{Re } \chi_R(g)}{d_R}\right) w_a(g) dg$$

exists and is finite. Then every face of area A has the limiting coefficients $e^{-A\psi(R)}$, and ψ has the form

$$\psi(R) = \frac{\sigma^2}{2} C_2(R) + \int_{G \setminus \{e\}} \left(1 - \frac{\operatorname{Re} \chi_R(g)}{d_R}\right) \nu(dg),$$

with $\sigma^2 \geq 0$ and ν a conjugation-invariant symmetric measure satisfying

$$\int_G \min(1, d(e, g)^2) \nu(dg) < \infty.$$

The limiting semigroup is Yang–Mills with $\lambda_2 = \sigma^2$ when $\nu = 0$ (including the degenerate zero-diffusion case), and $\nu = 0$ holds when, for every $\varepsilon > 0$, $a^{-2} \int_{d(e, g) > \varepsilon} w_a dg \rightarrow 0$.

Proof. Since $\hat{c}_R(a) \rightarrow 1$, $(A/a^2) \log \hat{c}_R(a) = -(A/a^2)(1 - \hat{c}_R(a))(1 + o(1)) \rightarrow -A\psi(R)$. Split the integral defining ψ at $d(e, g) = \varepsilon$. Near the identity, the invariant quadratic trace form is $1 - \operatorname{Re} \chi_R(e^X)/d_R = C_2(R)|X|^2/(2 \dim G) + O(|X|^4)$. Simplicity makes that trace form a scalar multiple of the chosen metric; conjugation invariance alone would not justify averaging over all directions in a general Lie algebra. This part converges to $\frac{\sigma^2}{2} C_2(R)$ with

$$\sigma^2 = \lim_{\varepsilon \rightarrow 0} \lim_{a \rightarrow 0} \frac{1}{a^2 \dim G} \int_{d(e, g) < \varepsilon} |X|^2 w_a(g) dg, \quad g = e^X,$$

along a subsequence if needed. Away from the identity, the measures $a^{-2} w_a dg$ restricted to $\{d \geq \varepsilon\}$ have bounded mass (a finite sum of the functions $1 - \operatorname{Re} \chi_R/d_R$ over representations separating the points of G is continuous and positive off e , hence bounded below there), so a subsequence converges weakly to ν on that set, conjugation invariant and symmetric. The integrability of ν near e follows from the bound on the near part. The form of ψ agrees with Hunt’s classification of convolution semigroups on Lie groups, restricted to central ones (Hunt 1956, metadata). A finite faithful representation controls both mass away from e and the truncated second moment near e , so compactness and the uniqueness of these limiting Fourier coefficients produce a probability convolution semigroup. Its continuity at area zero follows first on finite character sums and then, by uniform approximation of central continuous functions, weakly on G . The Lindeberg condition sets the far part to zero. Under these finite-exponent hypotheses the converse holds as well: if $\nu = 0$, testing away from e shows that the scaled tail masses vanish. $\square$

For general compact connected G , the diffusion term is $q_Q(R) = (2d_R)^{-1} \sum_{ij} Q_{ij} \operatorname{tr}(T_i T_j)$, where $dR(E_i) = iT_i$ and $Q \geq 0$ is an $\operatorname{Ad} G$ -invariant covariance. There is one scalar on each simple ideal and an arbitrary covariance on the central torus. For example $G = U(1)^2$ permits $q_Q(n_1, n_2) = (q_1 n_1^2 + q_2 n_2^2)/2$ with $q_1 \neq q_2$. Inversion symmetry removes drift and symmetrizes the jump measure. The single $\sigma^2 C_2/2$ formula therefore needs the simple-group restriction or an additional isotropy assumption.

Conversely, for any symmetric central Lévy semigroup μ_A , take $w_a dg = \mu_{a^2} * (k_{r_a} dg)$ with $r_a > 0$ and $r_a/a^2 \rightarrow 0$. These are smooth symmetric central densities and their limiting exponents are those of μ_A . This proves realizability of the entire stated class. The classification concerns independent plaquette refinement and its holonomy laws; additional regularity axioms for a particular continuum field theory need separate verification.

The limiting objects are Lévy’s two-dimensional Markovian holonomy fields, which he constructs from Lévy processes on G (Lévy, *Astérisque* 329, metadata). Theorem 2 adds only the lattice-side statement: which plaquette actions lead to which member.

Examples. The Wilson action $w_a \propto e^{\beta_a \operatorname{Re} \operatorname{tr} U}$ with $\beta_a \propto 1/(\lambda_2 a^2)$ concentrates like a Gaussian of width $\sqrt{\lambda_2} a$, satisfies the Lindeberg condition, and gives Yang–Mills. A weight carrying an explicit centre-vortex fugacity, $w_a = (1 - \kappa a^2) k_{\lambda_2 a^2} + \frac{\kappa a^2}{2} [k_{\lambda_2 a^2}(z \cdot) + k_{\lambda_2 a^2}(z^{-1} \cdot)]$ with z a generator of the centre $\mathbb{Z}_N$ of $SU(N)$, gives $\nu = \frac{\kappa}{2}(\delta_z + \delta_{z^{-1}})$ and

$$\psi(R) = \frac{\lambda_2}{2} C_2(R) + \kappa \left(1 - \cos \frac{2\pi k(R)}{N} \right),$$

with $k(R)$ the N -ality. On a spatial circle of circumference L the refinement note's eq. (10) becomes $E_R - E_0 = \hbar c L \psi(R)$: the vortex gas adds an N -ality-dependent string tension. The limit is action-dependent exactly through (σ^2, ν) .

Referee verdict, 2026-09-27: ACCEPT. The example is a probability mixture for $\kappa a^2 \leq 1$. Its coefficient is $e^{-\lambda_2 a^2 C_2(R)/2} [1 - \kappa a^2 (1 - \cos(2\pi k(R)/N))]$. For Euclidean time T the cylinder area is cLT , giving the stated energy of the circle transfer Hamiltonian with vacuum energy subtracted. At $\lambda_2 = 0$ neutral representations can have zero energy too.

3. Any dimension: the error budget of one step

Let t_n be the plaquette heat time at refinement step n , which by eq. (1) of the series/parallel note scales as $\lambda_D a_n^{4-D}$ at fixed physical coupling, and let L be a fixed physical box side. One directional step leaves three kinds of error.

- **Perturbative.** Coupling shifts and irrelevant operators of relative size $O(t_n)$ (Hypothesis P(α) and Proposition 7 of the series/parallel note).
- **Large-field.** Configurations far from the identity at the lattice scale, with action at least c/t_n , of density per physical volume about $a_n^{-D} e^{-c/t_n}$; here c depends on the lattice action (for heat-kernel $U(1)$ it is the vortex constant of Theorem 5 there).
- **Jump.** The action's Lévy content, the part of ν in Theorem 2.

These scales become a sufficient error budget only when they bound normalized measures or specified observables uniformly, with stability under blocking. In $D \geq 3$ plaquette constraints prevent inferring a Lévy classification directly from the two-dimensional theorem.

D	t_n , fixed λ_D	Perturbative	Large-field, $\sum (L/a_n)^D e^{-c/t_n}$
2	$\lambda_2 a_n^2 \rightarrow 0$	summable	summable; limit set by (σ^2, ν)
3	$\lambda_3 a_n \rightarrow 0$	summable	summable faster than any power
4	$g^2(a_n) \rightarrow 0$, logarithmically	resums into running	$\propto (a_n \Lambda)^{2b_0 c - 4}$, needs $2b_0 c > 4$
> 4	$\lambda_D a_n^{4-D} \rightarrow \infty$	no small parameter	no small-field regime

The perturbative column, with its sources. For $D < 4$, t_n falls geometrically along the dyadic sequence, so coupling shifts of relative order t_n sum to a finite renormalization: superrenormalizability. For three-dimensional lattice Yang–Mills, Balaban's ultraviolet stability is the constructive form of this statement in his block-averaging scheme (Balaban 1985, metadata). For $D = 4$ the heat time is the coupling itself, and each isotropic step shifts g^{-2} by the same amount, $-2b_0 \log 2$ at one loop, with $b_0 = 11N/(48\pi^2)$ for pure $SU(N)$ (Gross and Wilczek 1973; Politzer 1973; metadata). These shifts do not sum; they are the running coupling. Different lattice actions share the first two coefficients of the running and differ by a finite factor in Λ , computed at one loop for the Wilson action by Hasenfratz and Hasenfratz (1980) (abstract as indexed) and Dashen and Gross (1981) (metadata); for pure $SU(3)$ with the Wilson action the standard value is $\Lambda_{\overline{\text{MS}}}/\Lambda_{\text{lat}} \approx 28.81$ (Capitani 2003, review, as indexed). The heat-kernel action as a lattice action is treated by Menotti and Onofri (1981) (metadata). For $D > 4$, t_n grows as $a \rightarrow 0$ at fixed λ_D ; lattice Monte Carlo finds a phase transition between the confining strong-coupling regime and a weak-coupling spin-wave phase for $SU(2)$ in five dimensions, and none in

four (Creutz 1979, abstract). A continuum limit there would need a non-trivial ultraviolet fixed point, and none is established.

The four-dimensional threshold. With $g^{-2}(a) = 2b_0 \log(1/(a\Lambda))$, the convention of the refinement note §6, $e^{-c/g^2(a)} = (a\Lambda)^{2b_0 c}$. For pure $SU(N)$, $b_0 = 11N/(48\pi^2)$, so this particular leading-power expression for density per physical volume vanishes iff

$$c > \frac{2}{b_0} = \frac{96\pi^2}{11N} : \quad c > 43.1 \text{ } (SU(2)), \quad c > 28.7 \text{ } (SU(3)).$$

Correction (GPT-6 Astra referee), 2026-09-27 — REFINE: ACCEPT the strict power threshold; an action lower bound supplies only an upper density bound after entropy control, and equality requires prefactors and higher-loop terms.

For a dyadic sequence and a bound $Ca^{-4}[\log(1/a\Lambda)]^p e^{-c/g^2(a)}$, with fixed p and $g^{-2} = 2b_0 \log(1/a\Lambda) + O(\log \log(1/a\Lambda))$, the strict inequality ensures summability regardless of these logarithmic powers. At equality, an actual $n^{-p'}$ bound tends to zero if $p' > 0$ and is summable if $p' > 1$. An upper estimate that grows below threshold proves no survival or divergence. General defect families also need multiplicity, size-modulus and interaction bounds.

A continuum instanton has $c = 8\pi^2 \approx 79$, giving the density $(a\Lambda)^{11N/3} a^{-4}$ of small instantons, which vanishes for every $N \geq 2$ because $11N/3 > 4$. Lattice actions admit dislocations, small configurations of topological charge whose action lies below $8\pi^2$. In units of the instanton action the threshold reads $c/(8\pi^2) > 12/(11N)$, which is $6/11$ for $SU(2)$: this is the known dislocation criterion, found for the topological susceptibility by Pugh and Teper (1989) (abstract: with the plaquette action the $SU(2)$ susceptibility diverges in the continuum limit) and Gökeler, Kronfeld, Laursen, Schierholz and Wiese (1989) (metadata), and stated in the form “smaller than $6/11$ (for $SU(2)$) times the continuum value of a one-instanton configuration” by DeGrand, Hasenfratz and Zhu (1996) (passage). Suitably restrictive admissibility conditions can exclude specific rough dislocations; the precise action and cutoff must be given (Lüscher 1982, metadata, for lattice topology). Applying the power criterion to another defect family requires its own uniform density estimate. Higher-loop and determinant factors matter particularly at the equality threshold.

Correction (GPT-6 Astra referee), 2026-09-27 — REJECT the unconditional universality inference: the three scales alone do not prove a common continuum theory.

What this says about the mass gap. If normalized blocking estimates, iteration stability, large-field entropy control and continuum reconstruction identify a common limit, its physical spectrum can then be compared across regularizations. The expected dimensional forms are $C_3 \hbar c \lambda_3$ in $D = 3$ and $C_4 \hbar c \Lambda$ in $D = 4$. Proving the required common limit and its positive infrared threshold remains part of the mass-gap map. The present error budget identifies candidate sufficient estimates for that work.

4. Consequence for STATE

Item 1 of STATE gains its action and dimension dependence. The next steps on this line are to state Hypothesis P(α) for a general symmetric class-function action (second moment, large-field constant and jump content as its data), and to connect the four-dimensional large-field threshold to the H1 blocking hypothesis of the conditional theorem. For the joint paper, Theorem 2 gives the two-dimensional row of the comparison for every action at once, and the table gives the other rows.